

AIR COOLER INSTRUCTION MANUAL

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C O N T E N T S

I DESCRIPTIONS

II STRUCTURES

III OPERATION

**IV MAINTENANCE AND INSPECTION
DURING OPERATION**

V CLEANING

**VI PURCHASE SPARE PARTS
FOR MAINTENANCE**

I . Description

The Donghwa air cooler is a heat exchanger cooling the air supplied diesel engine to the specified temperature for improvement the performance of engine. It is called fin & tube type heat exchanger and used for MAN B&W and SULZER MAIN DIESEL ENGINE.

The heat removed from air is carried away by the cooling water (sea water or fresh water). The cooling water flows on the inside of the fin-tube, while air flows along the outside of the fin-tube.

We call a cooling water side as Water Side and fin-tube side as Air Side. The cooling methods of water side are differential into F.W(Fresh Water) cooling & S.W (Sea Water) cooling, and Central cooling system.

II . STRUCTURES

Air cooler mainly consists of cooler insert and casing. Cooler insert is made up of tube bundle and water cover.

II-1. Tube bundle

The bundle has several multi-finned tube block arranged in the direction of the air flow. According the cooler capacity, the number of block and size of each tube block are decided.

Both ends of the finned tubes are installed in the tubesheets by tube expansion with mechanical tube expander. And special hydraulic ball expander is used for the preference adherence of fin and tube. Fin-tube block is immortally fixed at tubesheet so that is unable to take apart each other.

The two side plates serve for protecting the finned-tube and for reinforcing the cooler performance by preventing air by-pass.

The channel cover is securely bolted to tubesheet and all bolt is tag-welded, while the reversing cover is only supported by stud bolt with clearance for absorbing thermal expansion of tubes and stud bolt is fixed with wire.

At side plate, lifting bosses are welded on for eye bolt and these make easier handling of unit. The supporting bars bolted on side plate serve for supporting tube bundle and for preventing the damage from a abnormal vibration. And also all bolts are tag-welded to prevent unlocking.

II-2 WATER COVER

Water cover has water inlet & outlet nozzle, partition plate for controlling water pass, air vent & water drain connection and chemical cleaning. And there is two jack bolt to easily detach cover from bundle for cleaning. Two water covers, the channel and reversing cover, are tightly bolted to corresponding tubesheet with gasket. Here water inlet & outlet nozzle detached cover is called channel cover and the opposite cover is called reversing cover. When the cooling medium is sea water, there is zinc anode or soft iron on water cover inside to protect cover from corrosion. The corrosion protecting medium, zinc anode or soft iron, is connected as flange type to exchange it when it was wear away and is bolted with gasket to water cover.

III OPERATION

III-1 Cooling Water Side

The bundle is to be filled with water and at the same time the air vent valve is to be opened so that the air in the bundle can escape through it. There is no more air bubbles in the escaping cooling water (it means air vented completely and the bundle has been filled with water), close the air vent valve tightly. If air remains in the tubes, which will not only deteriorate the performance, but also may cause accidents during operation, so that special care is required. After the cooler is filled with water completely, it may be O.K to operate the air cooler. When the pressure and temperature reach the normal operating condition, confirm that there are no abnormalities in regard to bolts, nuts, and gaskets. If water should be leaking from the gaskets, retighten the bolts and nuts. Continuously water leak even after retightening, the gasket should be exchanged for a spare gasket.

During operation, do not exceed the permitted condition. If do exceed, contact us and consider the operation condition. If you don't, we don't take the responsibility of resulting problem.

III-2 Air Side

Make sure that the operation on the cooling water side is normal, and then the cooler is fed with air. When the operating temperature has been reached, check on the air leakage. The charged air temperature inlet and outlet of the cooler is checked up and the cooling water quantity is to be corrected, if required. After operation start

for the air side, once more confirm that the connections of the cooling water side and the air side are secure.

III-3 Operation Stop

When stop the operation, first stop the air side and then the cooling water side.

When stop period will continue several days, open the drain valve in the cover and drain all water from the cooler to prevent from corrosion and freezing during the winter.

III-4 Cooling Water Velocity

Cooling water velocity is limited in view of erosion, corrosion, and fouling so that do operate over the minimum velocity. And the best tube inside velocity is staying the recommended velocity range.

Material	Recommended velocity (m/sec)	Minimum velocity (m/sec)
Copper	1.0 ~ 1.5	1.0
Al-Brass	1.4 ~ 2.0	1.0
Copper-Nickel (9:1)	1.8 ~ 2.5	1.0
Copper-Nickel (7:3)	2.4 ~ 3.5	1.0
Steel	1.5 ~ 3.0	1.0

※Note : Please never operate at the minimum velocity for the prolonged periods to prevent causing serious fouling.

IV Maintenance and Inspection during operation

IV-1 Cooling Management

- (1) With use for a long time, various contaminations are deposited inside of tubes and fins. It cause to deteriorate the performance of the air cooler, so following items should be checked during operation

- Temperature difference between inlet and outlet side of air and water
- Pressure loss for air and water

When a pressure loss has been doubled of initial condition, execute washing and cleaning.

- (2) When running the engine at part loads as well as during maneuvering or

running in winter season, the temperature or the amount of cooling water passing through the air coolers should be regulated so that the air temperature at the outlet of the cooler does not drop too low. If the air is cooled down to a dew point temperature, considerable amount of condensed water will collect in the air receiver. If the charged air temperature is above the dew point, condensates will form only where the air comes in contact with surface whose temperature is below the dew point. At this time, it is advisable to control the cooling water source temperature by adjusting circulation valve but inevitably if the cooling water quantity for the air cooler is controlled, the outlet valve should be throttled.

IV-2 Electrochemical Corrosion Protecting Metal

When the cooling water is sea water, a protection metal is installed inside the cover of the air cooler to protect the cooler from corrosion.

In order to protect the electrochemical corrosion of the cooler, the anode plates are installed in the cover. When operation is continued after the protection metal has been used up, notable corrosion will occur for the cover in case of standard equipment or for the tubes and tube plates when the cover is equipped with anticorrosion rubber lining etc., so that this metal should be periodically and should be replenished as required. The anode plate has to be exchanged for new one regularly. Generally the consumption of the anode plate in new cooler is very fast. So the first exchanging has to be done within 3 months. After that time the exchanging intervals are decided in accordance with the degree of the consumption on the last inspection.

Care should be taken that the surface of the protection metal and the threaded part are free of oil, paint, etc.

IV-3. Cover Coating Management

At the inside of the cover is coated with tar epoxy resin, neoprene lining, etc to protect the cover from corrosion by the cooling water, the inside coating should be inspected when the cover detached for inspection, cleaning, etc. If there is any damage, it should be repaired immediately. If you don't, local corrosion will occur at the damaged part, and finally the cover will be destroyed.

And also when you detach the cover, careful handling is required so that the surface will not be damaged.

IV-4. Use of the plug

When the tubes are broken because of corrosion or for some other reason, use the accessory plugs to plug both ends of the tube. The plug should be installed according to the followings

- (1) Drain the water completely from the cover.
- (2) Detach the cover installation bolts and nuts, and detach the cover. When it is difficult to detach the cover from the casing, the jack bolts should be used.
- (3) Drive the plug into both ends of the broken tube.

IV-5 Inspection

Normal hydro test pressure of water side is 6Kg/cm^2 . And air side test is done during operation whether leak or not. and general test pressure is 3Kg/cm^2 .

The holding time of air side and water side pressure test is 30 minutes. Only clean fresh water is used for the water pressure test of the cooler. Add soluble ferrous sulphate($\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$) to stimulate generating the protective film at new cooler. The quantity of soluble ferrous sulphate is 1ppm per hour.

It takes a time to be generating the protective film at new cooler tubes. So it is very harmful directly to contact tube with dirty harbour water. If dirty harbour water contact, the dirt film immediately generate on tube and will hinder the build-up of a protective layer.

V Cleaning

Use the cooler for a long time, various contaminations are deposited inside of tubes and fins. It cause to deteriorate the performance of the cooler. Check the cooler outlet air temperature and if it is increased considerably , and water side pressure loss is doubled when compared initial condition, you must execute cleaning.

Water side cleaning of the bundle should be carried out, if a considerable increase in air temperature measured at the outlet of the cooler. Keep in mind that a slight fouling at the water side is too much influencing to air cooler performance. Air side cleaning of the bundle should be carried out if the pressure loss of the cooler has

been about double of the initial condition.

V-1. Water side Cleaning

(1) Use Brushing Tools

- ① Turn off the blower
- ② Close the cooling water supplying valves and discharge pipe valves
- ③ Open the drain valve so that the bundle is to be empty.
- ④ Disconnect pipelines, and take off the both water covers using the jack bolts, if necessary.
- ⑤ If the unit comprises zinc protection plates or soft iron, these should be examined for their condition when water covers are detached. Damaged or wore down zinc plates or soft iron have to be exchanged new ones. The cover has to be sandblasted and given a new coat of tar epoxy, if the damaged cover area is large. Covers made of bronze or gunmetal don't need protective coats of paint and zinc protection plates but at that case soft iron is usually used for generating protective film on the tube surface.
- ⑥ Using the provided nylon brush tool, clean all tubes while they are still wet.
- ⑦ Rinse each tube with water to remove the brushed off dirt.
- ⑧ And then it is make sure that the tube inside is clean, and replace the covers and gasket after checking up the water side connections again.
- ⑨ After completely reassembled, operate water pump and do hydrotest to check leaking.
- ⑩ All work is done, open valves and operate the cooler.

(2) Chemical cleaning

When it is impossible to remove the fouling with tube brush, it is effective to circulate chemical cleaning solution. But, it take a considerable cautions, because if chemical cleaning fails, the tubes and others will be damaged, resulting in shortening the life of apparatuses remarkably. After finishing chemical cleaning, circulate fresh water so that cleaning solution is not remain in the cooling water. If you have no experience in chemical cleaning, consult the manufacturer.

Chemical cleaning should be carried out by circulating method as built-in or by dipping the bundle into some cleaning solution after pull out it from the casing

or between the air ducts. Working methods are following

- ① Turn off the blower
- ② Close the cooling water supplying valves and discharge pipe valves
- ③ Open the drain valve so that the bundle is to be empty.
- ④ Mix the chemical solution with water to specific concentrate, and sufficient amount as calculated water side volume.
- ⑤ Connect circulation pump and chemical tank at chemical connection on water cover. And then run pump and circulate solution.
- ⑥ When chemical cleaning has been done, stop pump and drain the solution through drain valve.
- ⑦ After solution draining, add neutralizer at chemical tank and neutralizing the remained solution.
- ⑧ Finish all of above steps, check again and run the cooler.

V - 2. The Air Side Cleaning

- (1) At moderation condition, it is possible to clean out by washing with hot water and compressed air.
- (2) But there are some deposits such as oil, grease, carbon, etc., it is preferable to clean it with steam blasts(3~7 kgf/cm², 150℃ of less) or hot water and compressed air.
- (3) It is impossible to clean by above, chemical cleaning must be carried out. Detach the bundle from the casing and dip it in cleaning solution. Chemical cleaning method is available as built-in by circulation of solution, when the cleaning system is equipped on board. After the bundle has been cleaned, it should be thoroughly flushed, dried, and properly reassembled.
Consideration should be taken to choose the chemical solution and cleaning process and must follow the guidance of chemical supplier not to attack tube or fin material.
- (4) When all work is done, you should add neutralizer and dissolve the remaining solution

V-3. Recommended chemical agent and supplier

	Chemical	Supplier
FOR WATER SIDE	SAF-ACID	= Drew Ameroid Japan Co., Ltd. Minato Ise Bldg. 12-1, 3-Chome, Kaigan-Dori, Naka-ku, Yokohama-si, Xanagwa-Ken, 231, Japan Phone : 045 - 212 -4741 Faxsimile : 045 - 212 - 4754 = Drew-Chemical
FOR AIR SIDE	ACC9	= Drew Ameroid Japan Co., Ltd. = Drew Chemical Co. New York, USA

※ Please consult chemical supplier's description.

VI Purchase Spare Parts for Maintenance

During operation period, exchanging from old parts into new ones is necessary for maintenance. For purchase, you know the detail structures and if necessary, refer to section II Structures and attached drawing.

The air cooler's parts names are tube bundle, water cover, gasket or packing, bolt, nut, anode, anode cover.

The ordering methods are following

- 1) See the nameplate and write down followings

SHIP NO., MANUFACTURE NO.

MANUFACTURING DATE, CLASS

If there is not name plate or hard to see, you can also find at left or right side of tubesheet.

- 2) Check 1) and numbers and names that you need parts.
- 3) Contact us by in person, mail, telephone, or fax and inform us.
- 4) Reconfirm your ordering list.

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